CSE 6179 – Digital Forensics

Hands-on Lab # 1 Data Acquisition (Disk Imaging Techniques) Spring 2026

1 Theoretical Background

1.1 Digital Forensics

Digital Forensics is a branch of forensic science that focuses on identifying, acquiring, processing, analysis and reporting on data stored on a computer, digital devices or other digital storage media. The computer, digital device or other digital storage system which holds valuable data for investigation is known as electronic evidence. Such examples are laptops, Smartphone, servers, Digital Video recorders, CCTV systems, drones, GPS systems, and game consoles. The following figure shows some digital devices.

Figure1. Digital Devices

The main goal of digital forensics is to extract data from the electronic evidence, process the data into useful information and present the findings for prosecution. All processes involved, therefore, should utilize sound forensic techniques to ensure that the findings are admissible in court. The nature of the cases in which digital evidence is involved is generally borderless and the offence happens in a split second; the findings derived from electronic evidence must therefore follow a standard set of guidelines to ensure that it is admissible not only in a specific country's court of law, but also in the international criminal justice system.

1.2 Understanding Electronic Evidence

Electronic evidence, unlike any other evidence, such as fiber, hair and gunshot residue, is challenging to process due to the fact that:

- ✓ The data can be scattered in several physical locations, sometimes across countries;
- ✓ The data can be transferred across jurisdictional borders effortlessly and in a matter of seconds;
- ✓ The data are highly volatile – easily altered, overwritten, damaged or destroyed by the single stroke of a key;
- ✓ The data can be copied without degradation;
- ✓ The lifespan of electronic evidence, unlike any other discipline of forensic evidence, is short before it is rendered useless. An example of this is a Smartphone: after five years, it may not be able to switch on or function properly.

Based on these facts, therefore, electronic evidence must be processed and handled with due care.

1.3 Principle of Electronic Evidence

When dealing with electronic evidence, the following principles must be adhered to:

- ✓ Electronic evidence must be obtained in a legal manner.

- ✓ The Staff involved must complete the appropriate training program, prior to handling electronic evidence.
- ✓ Any actions taken on the electronic evidence must not change its data. If it is necessary to access the original data or change the system setting, it is recommended that only competent staff be able to do so, and that staff must be able to justify those actions.
- ✓ Any action that requires the original data to be accessed or changed should be recorded and witnessed by a fellow practitioner if possible.
- ✓ A record of all actions taken when handling electronic evidence must be created and preserved so that they can be audited. An independent third party should be able to repeat those actions and achieve the same results.

1.4 Analysis Procedure

There are typically **four phases** involved in the analysis of electronic evidence in the Digital Forensics Laboratory:

- ✓ Acquisition
- ✓ Examination
- ✓ Analysis and
- ✓ Presentation.

Throughout the process, the chain of custody of the evidence must always be updated whenever it changes hands and its integrity must be secured at all times. Examination and analysis phases may be repeated until the work satisfies the case request.

It is commonly understood that conducting DF work in the DFL requires these four phases, however, not all cases will require all the phases. In certain cases, the acquisition phase can be skipped to conduct triage straightaway during the examination phase. An example of such a case is when there are large sets of data, where conducting acquisition on each evidence item may be not feasible. The following figure shows the laboratory analysis model:

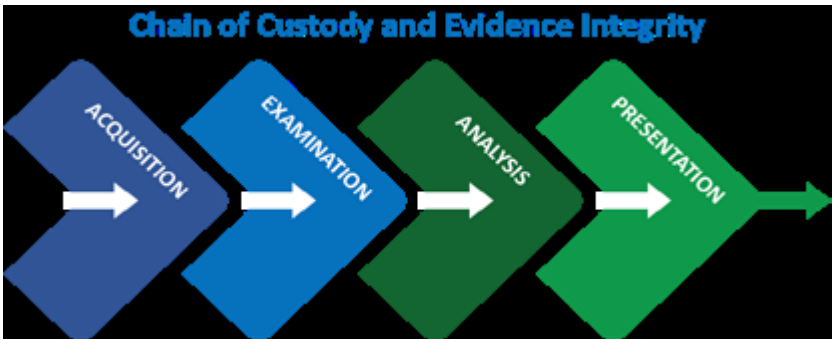


Figure2. Digital Forensics Laboratory Analysis Model

Acquisition or, as it is better known, data acquisition, is the process of creating a forensic copy of the electronic evidence (exhibit) such as hard disk, thumb drive or server in the form of an image file or files. The image file or files will then be used for the next stage of the process in analysis the evidence. The acquisition is made in order to preserve the integrity of the electronic evidence. It is to produce an identical copy of the data without changing the content of the electronic evidence in any way.

Electronic evidence needs to be acquired in a forensically sound manner. Data is typically acquired by collecting volatile data from a running computer during a search, or by acquiring a storage medium from a seized computer or at any other stage during an investigation. The intangible nature of data and information

stored in electronic form makes it easy to manipulate and more prone to alteration than traditional forms of evidence. It is therefore important to have a defined and tested acquisition procedure.

Once an image file has been created, both the hash value of the exhibit and the image file must be recorded. Hashing is used to prove that the image file is exactly the same as the content of the exhibit. There are a lot of hashing algorithms used in DF, such as Sha-256. Most forensic software and hardware offer the hash-generating feature.

Examination and analysis must only be done on a forensic copy of the original evidence, unless circumstances prevent examiners from doing so. This is important in order to preserve the integrity of the evidence. The forensic copy of the electronic evidence must be stored on other storage media, never onto the evidence itself. The forensic copy must be clearly labeled to ensure it is not mixed up with the original evidence or with forensic copies from other cases. The DFL must therefore prepare some storage media before receiving cases.

Types of Data Acquisition of computers:

There are two levels of data acquisition:

- ✓ Physical data acquisition and
- ✓ Logical data acquisition.

While physical data acquisition includes all raw data, a logical copy typically only includes an allocated subset of those data. Physical data acquisition, at whole disk level, copies all data contained on the disk, including the partition scheme, partitioned area, and un-partitioned area. Logical data acquisition on disk level copies only a logical partitioned area.

The Examiner commonly chooses physical data acquisition of a whole disk because it includes deleted files and unallocated clusters. When dealing with encryption, however, a logical data acquisition of unlocked data is preferred to a physical acquisition of the encrypted data. In this case physical acquisition is still recommended if the forensic software support the mounting of the encrypted imaged.

Physical Image: A physical image of a hard drive will capture all of the ones and zeroes contained on the drive. It will capture the deleted space on the hard drive even if the drive has been recently formatted. It will capture deleted files and file fragments on a hard drive. If one is making a physical image of a 1 TB drive the resulting image file(s) will be 1 TB, unless compression algorithms are used.

Logical Image: A logical image of a hard drive will capture all the “active” data. If you look at the My Computer icon on your computer and browse through the C drive you are viewing the logical drive and active files. This is what will be captured if one performs a logical capture. Typically, deleted space, deleted files and fragments will NOT be captured. If one is making a logical image of a 1 TB drive, but only 30 GB is active files, then the resulting image will be 30 GB uncompressed.

We'd always interpreted it this way:

- ✓ **Physical** = bit for bit of all partitioned and unpartitioned accessible space on the drive
- ✓ **Logical** = bit for bit of a logical partition (i.e. if a hard drive has 2 logical partitions then two images will be required). The logical will still include unallocated space that is part of that partition, the only parts it misses is unpartitioned or unpartitionable space.

To create a copy, the Examiner first needs to choose the state of the exhibit. If the system is up and running, the Examiner may need to choose live acquisition. However, if the system is powered off, the Examiner may then choose to conduct dead acquisition. The difference between these methods of acquisitions is described in the following table.

	Dead Acquisition	**Live Acquisition**
What?	Dead acquisition is conducted on a dead system. A dead system is a system that is not running; turned off, with no power. When the system is dead, volatile data in temporary storage areas such as RAM memory, running processes, cache or active application dialogues on a computer will no longer be available.	Live acquisition is conducted on a live system. A live system is a system that is up and running where information may be altered as data is continuously being processed. Because of the rich evidentiary value that could be discovered in a live system, switching it off may cause loss of volatile data, such as data stored on the cloud, encrypted data, running process, network connected and mounted file system.
How?	The process of conducting dead acquisition is straightforward as it is normally done automatically using forensic equipment. The hard disk must first be taken out of the computer before connecting it to the equipment, if possible. In some cases notebook computers or devices with soldered solid state drive storage cannot be extracted in dead acquisition. Other methods to perform extraction in such cases, like booting the system with a live CD/USB, should be considered.	Data on a system have different levels of volatility. These data will be lost if the system is switched off or rebooted. Whenever the Examiner acquires live data, it is sensible to collect from the most volatile data to the least volatile. The typical level of volatility, from the most to least volatile is as follows: ✓ Memory ✓ Swap File ✓ Network Processes ✓ System Processes ✓ File System Information
When?	Dead acquisition is conducted when: ✓ System is switched off ✓ Deleted data is more important than volatile data	Live acquisition is conducted when: ✓ The system is business-critical and cannot be shut down ✓ Volatile data are more important than deleted data

Table 1. Method for conducting acquisition - **Dead Acquisition** and **Live Acquisition**

The Examiner then needs to choose whether to clone the exhibit or to create an image. The clone copies the data **bit-by-bit** from one storage medium to another. The image, on the other hand, copies the data **bit-by-bit** from one storage medium into **an image file**. This file can then be stored on another medium. The latter technique is the **more commonly used** as the image file can subsequently be read by most forensic software for forensic analysis processing. Cloning is commonly used for simulation purposes. Before that can be done, the Examiner must ensure that the exhibit is **write blocked**; a method which enables only reading and prevents any writing on the exhibit.

1.5 Write Blocker

A write blocker is a device that enables data to be acquired from a hard disk without modifying the disk's data. The device allows a read command, but does not allow write commands to be executed on the hard disk. Most imaging tools have a **built-in write blocker** that the Examiner can utilize while imaging a hard disk. While write blocking can also been achieved by software tools or changes in the Windows registry, hardware solutions will be preferred in DFLs. The following figure is a **write blocker device**.

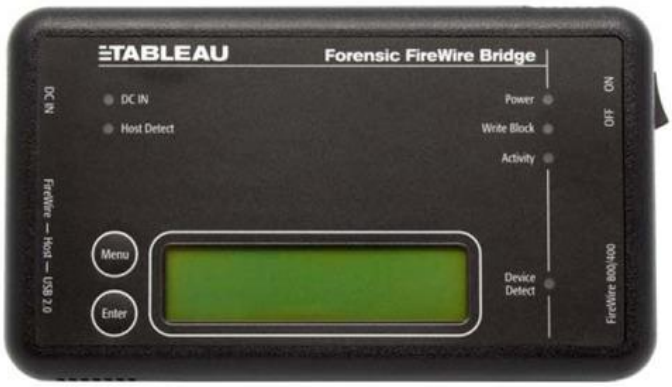


Figure 3. Write Blocker Device

Figure 4 shows a typical acquisition setup, where a hard disk of a computer is connected with a write blocker. Write blockers can take two forms: hardware or software (you can see a picture of a hardware write blocker here). The hardware devices are more reliable, stopping all write commands from reaching the digital media. Software writer blockers are less reliable and tend to be proprietary. Acquired media is usually referred to as an "image", they are stored in a number of open and proprietary formats. The popular EnCase software employs a proprietary, compressible, "EnCase Evidence File Format" (EEFF). Other open formats such as RAW (i.e. a simple bit copy) are used by programs such as "FTK Imager".

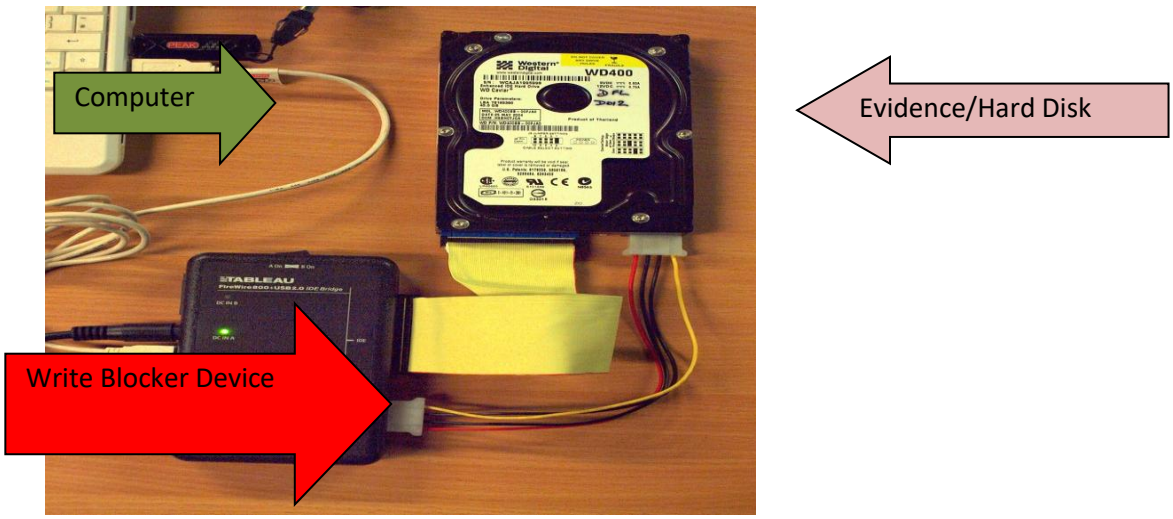


Figure 4. Example acquisition setup

1.6 Imaging Tools

The imaging of a storage medium can be performed using forensic software or hardware. There are free, as well as commercial, products available which can aid in the process. When purchasing a tool, the most important criteria to look for are the **speed of conducting the image** and the **reliability**. The imaging software can include features such as:

- ✓ recognition of hidden areas

- ✓ imaging multiple devices simultaneously
- ✓ imaging to multiple destinations concurrently
- ✓ hash verification with common hash algorithms
- ✓ hash verification at different stages of the imaging process
- ✓ support the most common forensic image formats
- ✓ producing encrypted and compressed images
- ✓ resuming an interrupted acquisition process
- ✓ tolerance of hardware errors

1.7 Imaging Format

There are several common image file formats, namely **raw** or **dd**. These formats store all data from the original medium in a raw file. Other formats include **Expert Witness Format (EWF)** and **Advanced Forensic Format (AFF)**. They contain features such as:

- ✓ Compression of data
- ✓ Encryption of data
- ✓ Error-Checks
- ✓ Case Metadata
- ✓ Hash sums
- ✓ Splitting the image in chunks

Furthermore, different forensic software solutions come with their own proprietary image formats with similar features. When choosing an image format, always opt for one that is supported by most forensic software solutions. Some DFLs use different forensic software, and as a result there is the possibility that the image file may not be opened if the Examiner chooses a unique image file format.

1.8 Process Flow

The common process for conducting data acquisition is illustrated in the following figure:

Figure3. Data Acquisition Process on Computer

The method of conducting data acquisition is described in the following section.

1.8.1 Identify Storage Media:

The Examiner must prepare a compatible storage media, with sufficient data size before handover. If the exhibit is large, the Examiner may need to prepare several storage media to store the image file.

1.8.2 Image the Exhibit:

To image the exhibit, first connect it to a write-blocker to ensure the exhibit is not writable, thus protecting its integrity. Most forensic tools offer this feature. The image file is then stored in the prepared storage media. To protect the storage media and the image file, use a standard labeling format by hashing all the evidence acquired with SHA256.

1.8.3 Verify Exhibit and Image File:

After the image file has been created, the Examiner needs to check that it is able to run using forensic software, and that the hash values of both the exhibit and the image file are matched.

1.8.4 Document All Actions:

The last step in examining and analysis the computer is to document the process, the tools used, the hash values, date and time, as well as the Examiner's initials in the case notes.

1.9 Computer forensic tools:

During the 1980s, most digital forensic investigations consisted of "live analysis", examining digital media directly using non-specialist tools. In the 1990s, several freeware and other proprietary tools (both hardware and software) were created to allow investigations to take place without modifying media. This first set of tools mainly focused on computer forensics, although in recent years similar tools have evolved for the field of mobile device forensics. This list includes notable examples of digital forensic tools.

Name	Platform	Description
Autopsy	Windows, macOS, Linux	A digital forensics platform and GUI to The Sleuth Kit
Magnet IEF Magnet AXIOM	Windows, Clouds and Mobile	Full digital forensics suite created by Magnet Forensics.
Belkasoft Evidence Center	Windows	Digital forensics suite created by Belkasoft
BlackLight	Windows macOS iOS Android	Digital forensics for major mobile operating systems
COFEE	Windows	A suite of tools for Windows developed by Microsoft
EnCase Forensic EnCase Imager	Windows	Digital forensics suite created by OpenText Software
Forensic Explorer	Windows	Digital forensics suite created by GetData
FTK Forensic FTK Imager	Windows	Multi-purpose tool, FTK is a court-cited digital investigations platform built for speed, stability and ease of use.
OSForensics	Windows	Multi-purpose forensic tool
Rocket Cloud	Windows	Digital forensics acquisition and case management platform created by Digital DNA

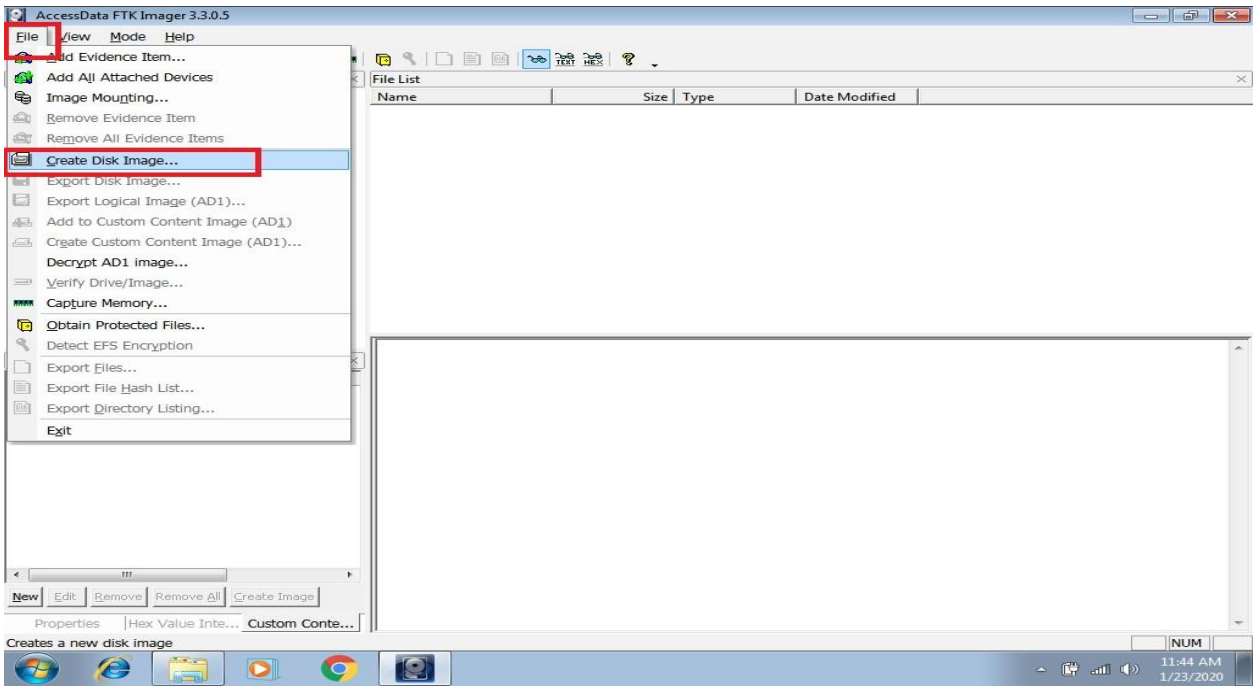
Name	Platform	Description
		Group
SafeBack	N/a	Digital media (evidence) acquisition and backup
SANS Investigative Forensics Toolkit - SIFT	Ubuntu	Multi-purpose forensic operating system
The Sleuth Kit	Unix-like/Windows	A library of tools for both Unix and Windows
UltimateForensics	Windows	Forensic software for image/video search and analysis by means of matching visual content
X-Ways Forensics	Windows	Integrated computer forensics environment created by X-Ways

2 Instructions

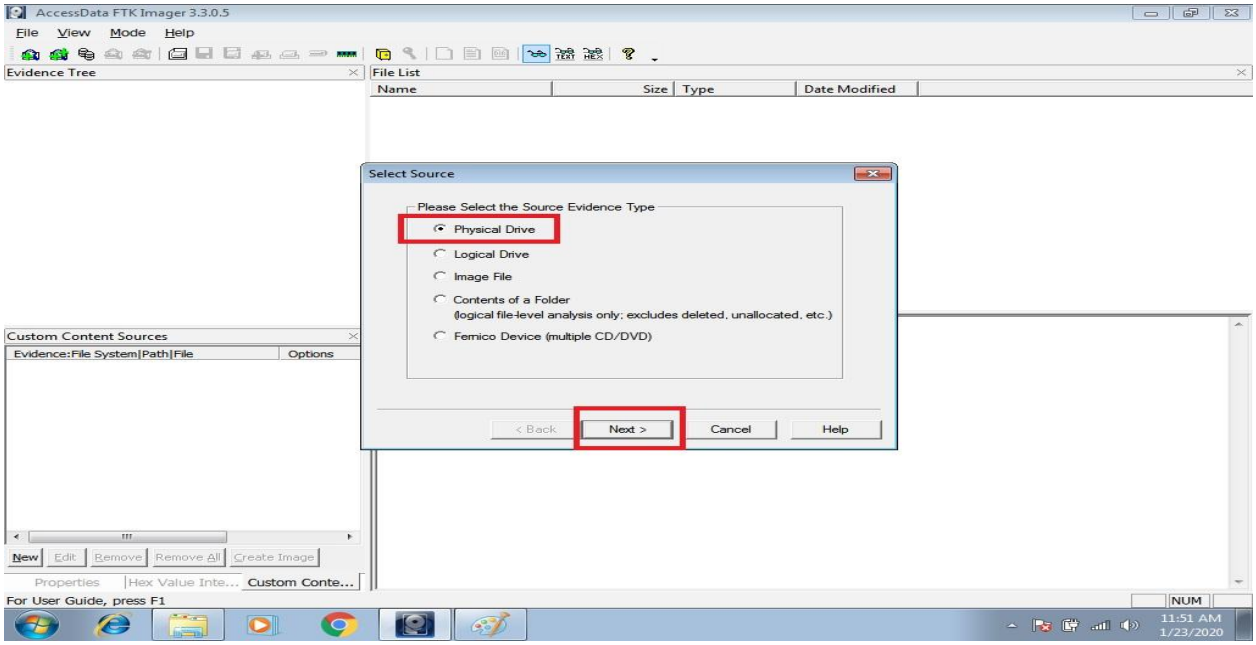
Forensic Toolkit or FTK is a computer forensics software product made by AccessData. This is a Windows based commercial product. For forensic investigations, the same development team has created a free version of the commercial product with fewer functionalities. This FTK Imager tool is capable of both acquiring and analyzing computer forensic evidence.

How to image of Hard Disk/Flash Drive by using "FTK Imager" Software

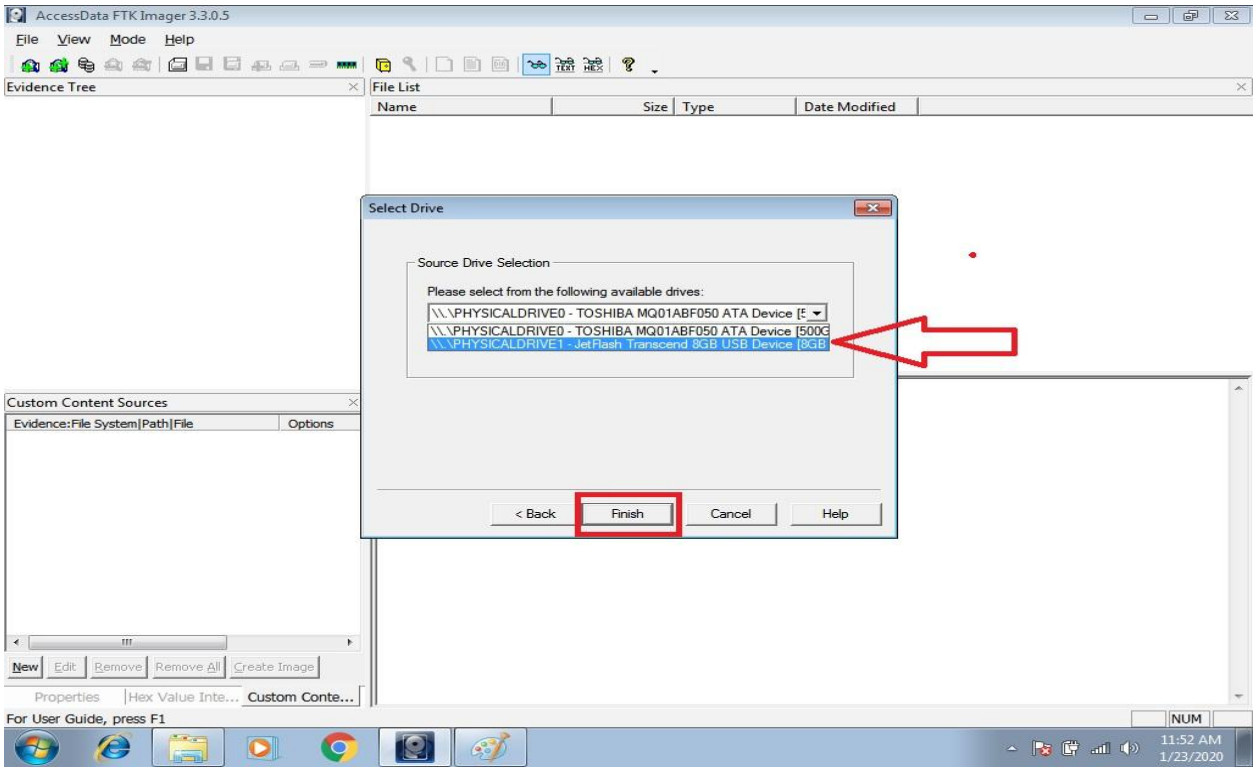
1. Click "File" from the menu bar then go to "Create Disk Image" and open it.



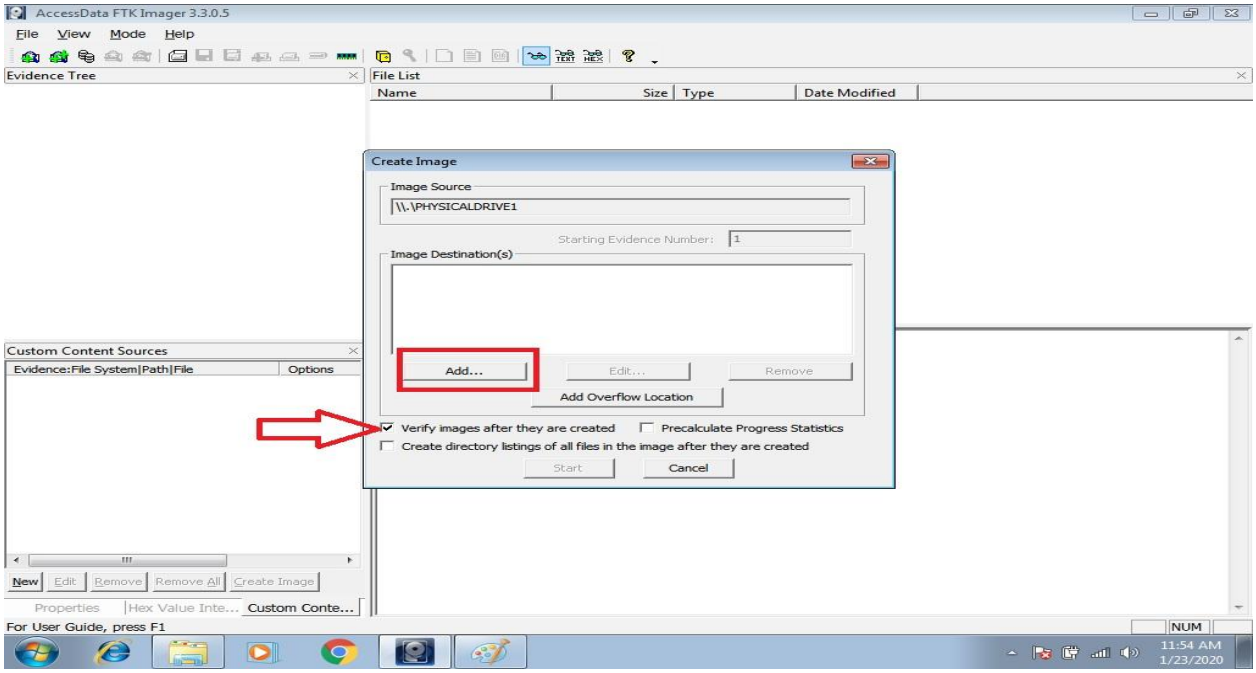
2. Select "Physical Drive" for imaging hard disk/flash drive which is already attached with PC then click "Next" button.



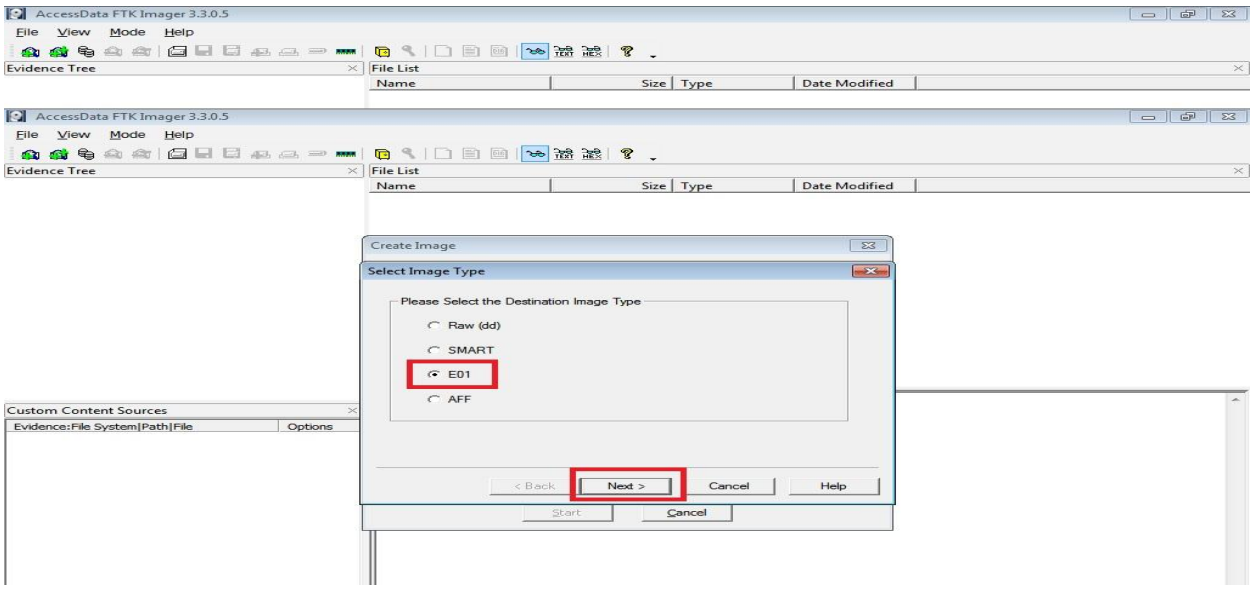
3. Select the target source drive from available attached drive/drives and click "Finish" button.



4. Check mark on "Verify images after they are created" then click the "Add" button.



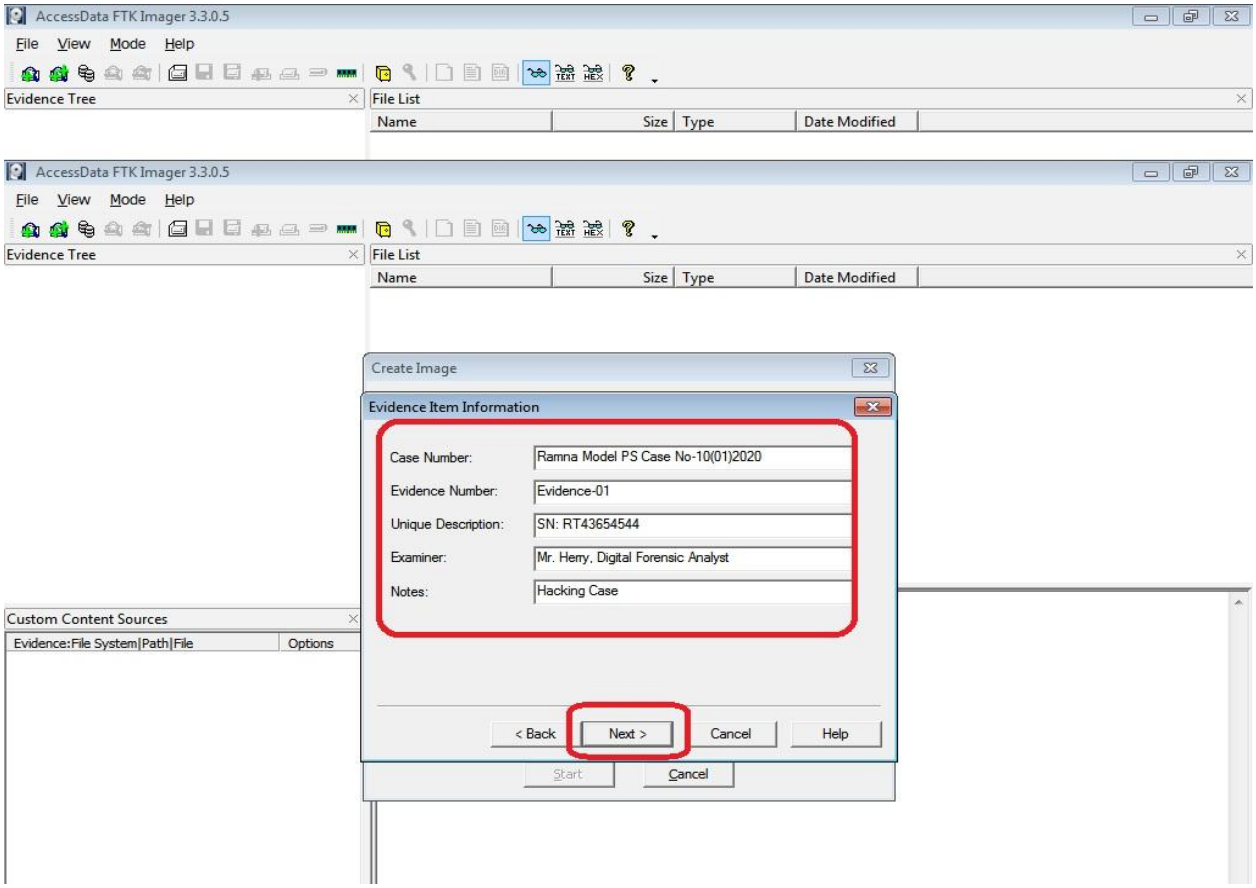
5. Select image file format like as "E01" (EnCase Image File Format) then click "Next" button



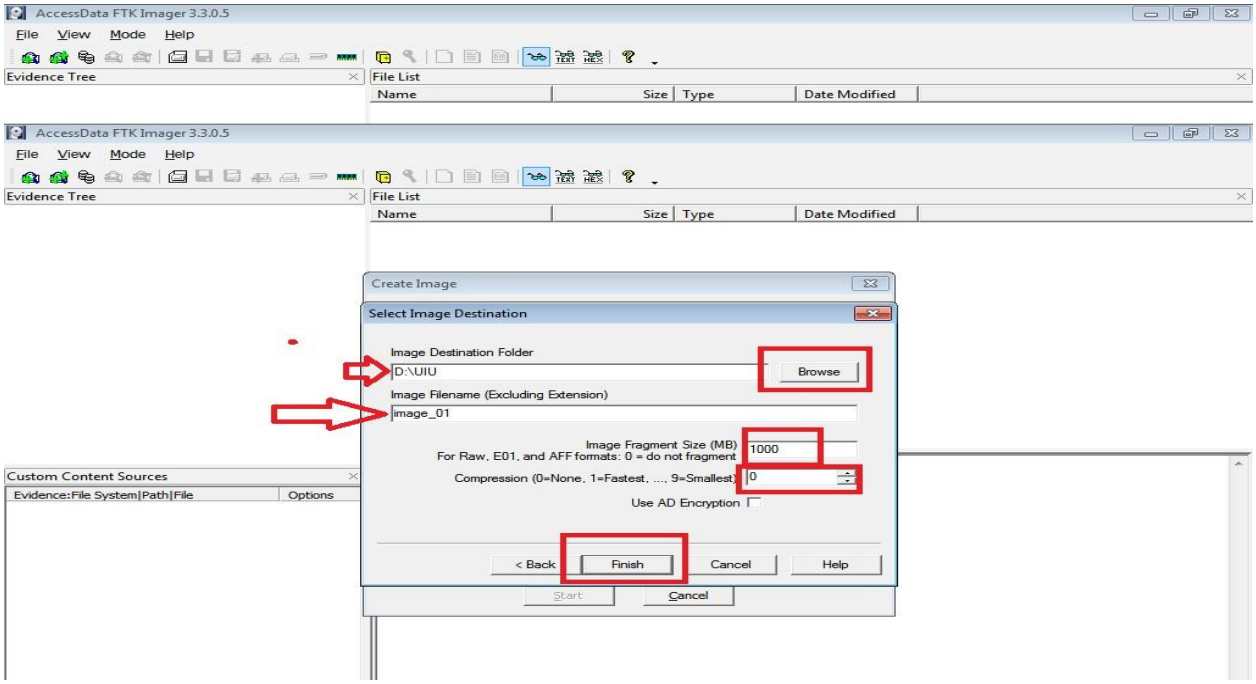
- ✓ **Raw (dd):** This is the image format most commonly used by modern analysis tools. These raw file formatted images do not contain headers, metadata, or magic values. The raw format typically includes padding for any memory ranges that were intentionally skipped (i.e., device memory) or that could not be read by the acquisition tool, which helps maintain spatial integrity (relative offsets among data).
- ✓ SMART: This file format is designed for Linux file systems. This format keeps the disk images as pure bit streams with optional compression. The file consists of a standard 13-byte header followed by a series of sections. Each section includes its type string, a 64-bit offset to the next section, its 64-bit size, padding, and a CRC, in addition to actual data or comments, if applicable.

- ✓ **E01:** this format is a proprietary format developed by Guidance Software's EnCase. This format compresses the image file. An image with this format starts with case information in the header and footer, which contains an MD5 hash of the entire bit stream. This case information contains the date and time of acquisition, examiner's name, special notes and an optional password.
- ✓ **AFF: Advance Forensic Format (AFF)** was developed by Simson Garfinkel and Basis Technology. Its latest implementation is AFF4. The goal is to create a disk image format that does not lock the user into a proprietary format that may prevent them from being able to properly analyze it.

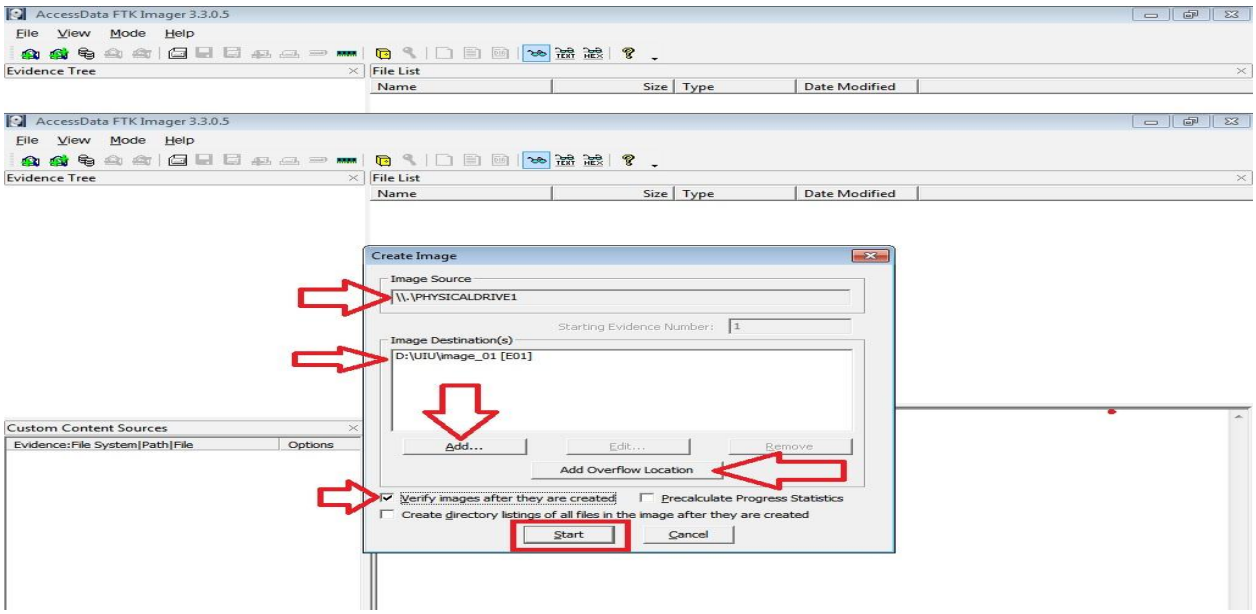
6. Fill up "Evidence Item Information" field according to your case details then click "Next" button.



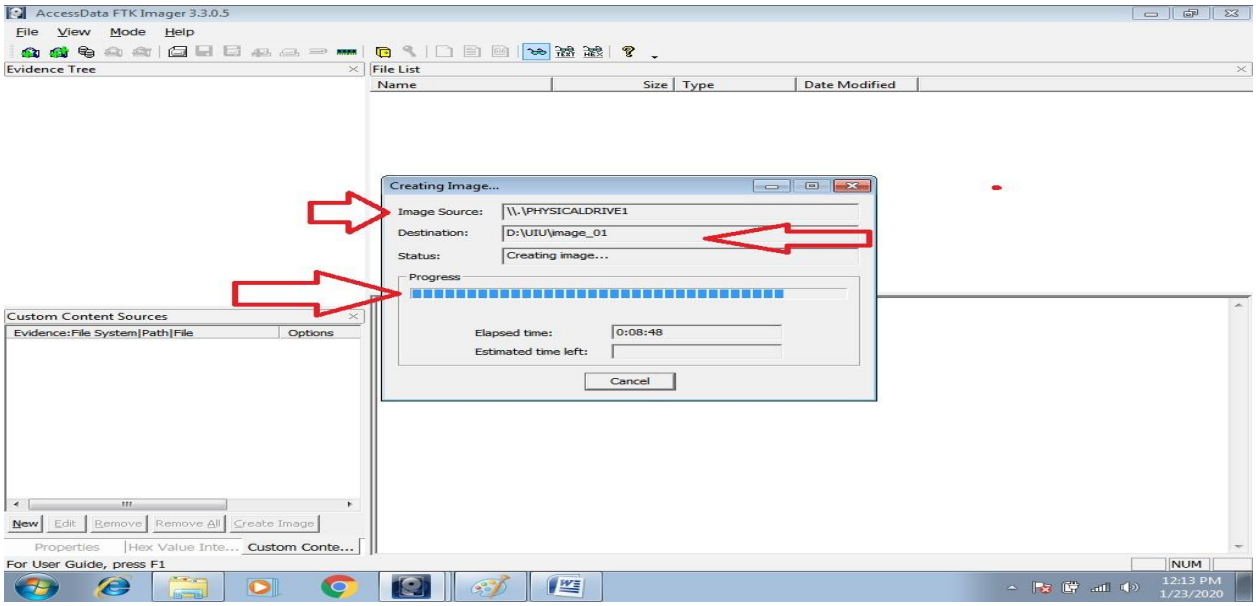
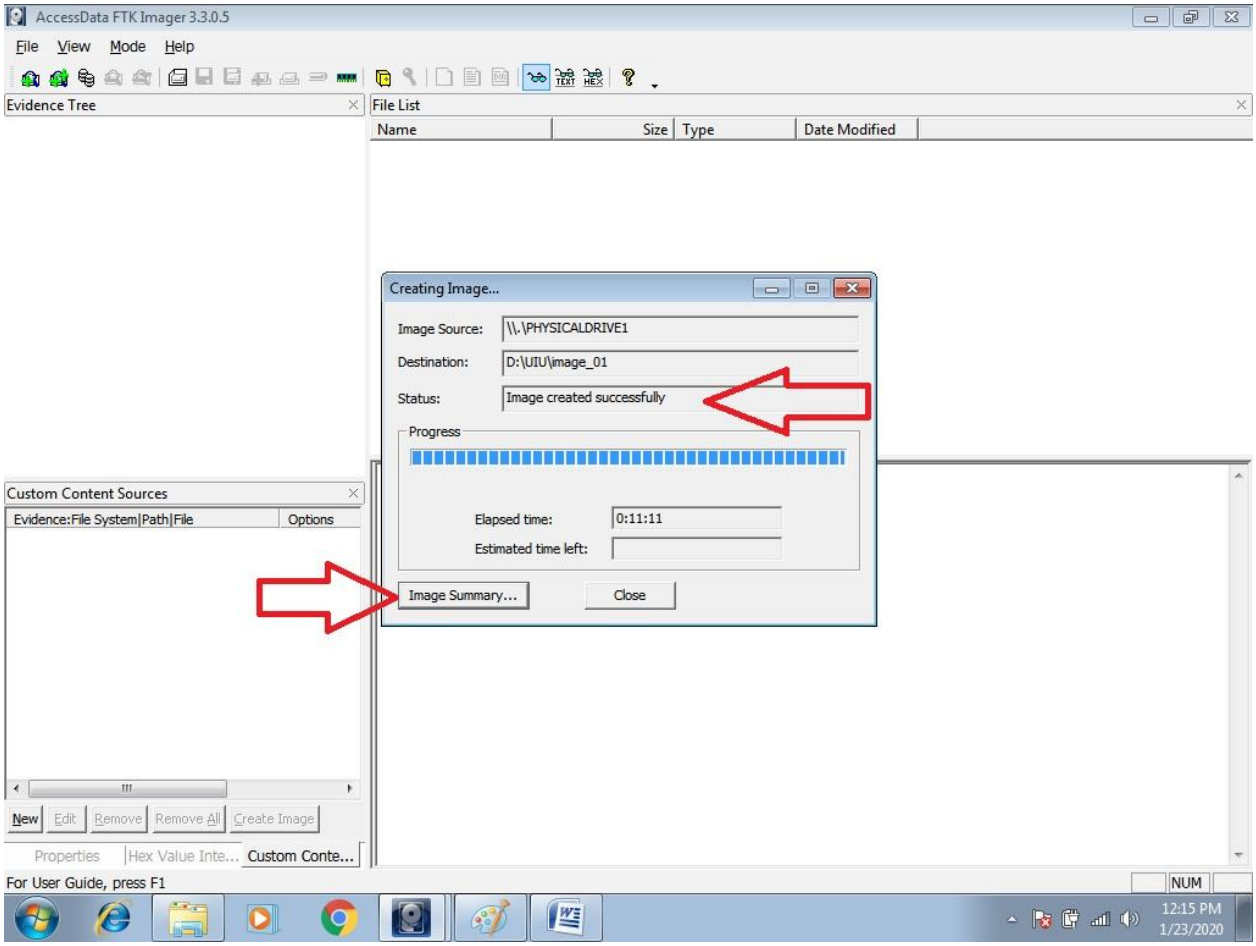
7. Select image destination folder and click "Finish" button.



8. Check your given settings if it's okay then finally click "Start" button for creating image.



9. The following picture is shown that the process is running.



10. When the image is completed, the status field shows "Image Created Successfully" then click "Close".